

Time's Up

...for a CPU upgrade? Here are 11 contenders for the slot

The mini-budget is out, and that means it's upgrade time! With this in mind, we try to find the perfect complement for the motherboards we compared in 'A Board's-eye View' on page 40.

One look at the market, and it's clear that options are limited to just Intel and AMD; VIA is out there somewhere, but its appeal is severely limited. Motherboards that use VIA processors are proprietary, with far fewer upgrade options. The performance of these systems can't be compared to that of the Pentiums or the Athlons. Therefore, we'll be concentrating mostly about AMD and Intel CPUs. We received and tested 11 processors in all.

A look under the hood

Intel Pentium 4 processors are based on the Northwood core (See box 'CPU Codenames Demystified'), with 512 KB of L2 cache. The Celeron, however, only has an L2 cache of 128 KB, and runs at 400 MHz Front Side Bus (FSB). It falls in the Value category.

Newer processors such as the Intel Pentium 4 3.2 GHz and the Intel Pentium 4 2.8C GHz use the same core technology. The difference is in the FSB, which has been pumped up in both processors to 800 MHz, effectively resulting in an increase in processing speed.

With the introduction of the 2.8C GHz processors, consumers have tasted Hyperthreading—a technology that enables one processor to execute two instruction sets at the same time. Each of these instruction sets can be independently halted or interrupted. To the OS, it appears that the system has two processors. Applications such as Adobe Photoshop and 3DStudio Max benefit greatly from this architecture, and have threading-specific optimisations.

All the processors in the Pentium 4 line-up are based on the same core; performance differences are due to increased core clock frequencies and pumped-up FSBs. These increased FSB speeds only give a slight performance increase if the clock speeds change marginally. It makes more sense to keep a Pentium 4 2.8C, rather

than getting a Pentium 4 3.0 GHz, which will cost you an extra Rs 3,200.

In the AMD line-up, we had four mainstream processors—the Athlon XP 2400+, the Athlon XP 2600+, the Athlon XP 2800+ and the Athlon XP 3200+. The Athlon XP 2400+ is based on the Thoroughbred core, and is fabricated using the 0.13 micron process. The Thoroughbred core supports features quite different from what Intel processors do. The fabrication process is the same, but there's a new thermal diode, along with a robust cooling solution. The Thoroughbred core also features 3DNow! Professional technology, with Streaming Single Instruction Multiple Data Extensions (SSE) compatibility. This lets it run applications such as games,

video -

encoding and multimedia that use the SSE instruction set.

With the introduction of the Athlon XP 3200+, AMD has introduced the new Barton core with an FSB of 400 MHz, while the core clock speed of the processor is still 2.2 GHz. The major difference is that the L1 cache has been increased to 128 K, and the L2 cache has been pumped up to 512 K. This gives the 3200+ a performance increase over its earlier Thoroughbred counterparts. In fact, the Barton core has now been moved to the AMD 2800+ models, and is available in the market. To identify a Barton Athlon XP 2800+, all you need to check is the amount of L2 cache memory, mentioned on the box—it should be 512 KB. AMD has also released a new version of the 2600+ with 512 KB of cache, but the processor still runs at 333 MHz FSB.

